Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 1 of 12 · Question 5 part c.i · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5 (a) (i) | | • alternative form of gene;<br>• only expressed in homozygote/when 2 recessive alleles are present/ when no dominant allele is present/not expressed in heterozygote/; | | 2 |
| (b) | | any two from<br>• person 2;<br>• person 4;<br>• person 9; | | |
| | (ii) | Aa; | | 1 |
| (c) | (i) | parents Aa × Aa;<br>gametes A a A a;<br>F1 AA 2Aa aa;<br>phenotype normal normal carrier sickle-cell; | 2 marks max for unlabelled Punnett square (mp's 2 and 3) | 4 |
| | (ii) | • chance of a boy = 0.5/50%;<br>• chance of sickle cell anaemia = 0.25/25%;<br>• chance of boy with condition = 0.5 × 0.25 OR $\frac{50 \times 25}{100}$;<br>• = 0.125/12.5%; | | 4 |

Total 13 marks

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 1 of 12 · Question 5 part c.i · 4 marks

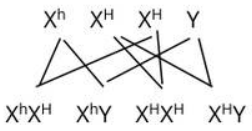
2023 NOV · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 2 of 12 · Question 9 part c.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 9 | (a) | (i) | A (the inheritance of two different alleles, both of which are expressed)<br><br>B is incorrect as one allele is not expressed in codominant inheritance<br>C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles<br>D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles | | 1 |
| | | (ii) | $I^AI^A$ and $I^AI^o$ | | 1 |
| | (b) | (i) | • blood group O contains no antigens;<br>• antigens removed/digested (by enzyme);<br>• carbohydrase digest carbohydrates;<br>• protease digest proteins; | | Max 3 |
| | | (ii) | reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection; | | 1 |
| | (c) | (i) | parent genotypes<br>$X^hX^H$ and $X^HY$;<br><br>gametes and linkage<br>$X^h$ $X^H$ $X^H$ Y<br><br>  <br> ; | One mark for parent genotypes and one mark for gametes **and** linkage | 1<br><br>1 |
| | | (ii) | 1 in 4/25%/quarter/1/4 /1:3/0.25; | | 1 |

Total question 9 = 9 mark

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 2 of 12 · Question 9 part c.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 3 of 12 · Question 8 part b.i · 4 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 8 | (a) | any three from<br>• change in base sequence/named mutation (1)<br>• of DNA (1)<br>• causing change in genotype/gene (1)<br>• incorrect protein produced (1)<br>• change in phenotype/characteristics (1) | Allow change in DNA for mps 1 and 2 (1) | 3 |
| | (b) (i) | father $X^HY$ mother $X^HX^h$ (1)<br>gametes $X^H$ Y $X^H$ $X^h$ (1)<br>F1 $X^HX^H$ normal female; $X^HX^h$ carrier female; $X^HY$ normal male; $X^hY$ haemophiliac male (1) | | 4 |
| | (ii) | (chance of first and second child is) ¼/0.25/25% (1)<br>for both is ¼ x ¼/0.25 x 0.25 (1)<br>= 1/16/0.0625/6.25% (1) | | 3 |
| | (c) | any five from<br>• males only have one X chromosome (1)<br>• two X chromosomes/H and h needed to be a carrier (1)<br>• recessive allele/h (1)<br>• carried on X chromosome (1)<br>• either present or not (1)<br>• no chance for dominant allele to be present (1)<br>• to mask recessive (1) | | 5 |

Total 15 marks

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 3 of 12 · Question 8 part b.i · 4 marks

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## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 4 of 12 · Question 8 part b.i · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 8 (a) | (i) | absorbs uv light;<br><br>reduces/prevents risk of skin cancer; | Allow protects against (UV) radiation | 2 |
| | (ii) | eye/ retina/ choroid/ red blood cell/ haemoglobin ; | | 1 |
| (b) | (i) | mother x father<br>Hh hh;<br><br>gametes H h h;<br><br>fertilisation Hh hh;<br><br>phenotype white forelock normal; | Reject diagrams that show sex linkage | 4 |
| | (ii) | chance of producing a boy is 0.5/½;<br><br>chance of producing offspring with condition 0.5/½;<br>chance of producing boy with condition is ½ × ½ = ¼; | Allow 50% or equivalent anywhere in calculation<br><br>Full marks for correct final answer | 3 |
| (c) | | Any 4 from<br>different (genetic) code produced;<br><br>different order of amino acids;<br><br>codes for different proteins/enzymes;<br>enzyme substrate complex not formed;<br>causes change in pigment; | Allow change in base sequence/genetic code<br>Allow any reference to change in amino acids | 4 |

Total 14 marks

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 4 of 12 · Question 8 part b.i · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 5 of 12 · Question 7 part b.ii · 6 marks

2022 JAN · Paper 02 · find part b.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 7 | (a) (i) | • appears in both males and females;<br>• in similar numbers; | | 2 |
| | (ii) | • condition present in heterozygous genotype/only one allele needed/affected are carriers;<br>• only some children (from affected and unaffected parents) are affected/not affected; | | 2 |
| | (b) (i) | • probability of a boy is 0.5/50%;<br>• probability of no Huntington's is 1.0;<br>• overall probability is 0.5 × 1.0 = 0.5/50%/1; | | 3 |
| | (ii) | person 3 must be hh/homozygous recessive;<br>does not carry a faulty allele;<br>person 2 is Hh/heterozygous/carries a faulty allele / carries one normal allele;<br>as father not affected;<br>offspring either Hh or hh;<br>half with condition and half without/50% chance of being affected; | marks can be awarded for suitable genetic crosses that are annotated | 6 |

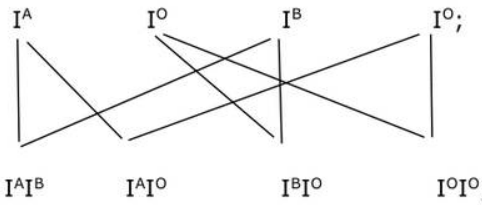
Total 13 marks

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 1 of 7 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (a) | (i) | **Africa**<br>Blood group / Percentage of blood group<br>O: 68<br>A: 17/17.5<br>B: 10;<br>AB: 5<br><br>**Europe**<br>Blood group / Percentage of blood group<br>O: 45<br>A: 40<br>B: 12<br>AB: 3; | | 2 |
| | (ii) | • reference to antigens on red blood cells (group B);<br>• antibodies produced;<br>• causing agglutination/clumping (of red blood cells); | | 3 |
| | (iii) | $I^AI^O$ $I^BI^O$;<br>  <br><br>$I^AI^B$ $I^AI^O$ $I^BI^O$ $I^OI^O$; | Full marks for a Punnett square if parent genotypes are given. | 3 |
| | (iv) | 1 in 4:/25%/¼/0.25; | | 1 |
| (b) | | • <u>deoxygenated</u> blood moves into the left side/ventricle of the heart;<br>• (deoxygenated blood) mixes with <u>oxygenated</u> blood;<br>• less oxygen travelling around the body/to body/cells/in bloodstream;<br>• breathes faster to get more oxygen into the body; | | 4 |
| | | | | Total 13 |

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 2 of 7 · find part a.iii in this mark scheme extract

| | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 3 of 7 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| 6 | Any six from the following:<br>• remove plasmid from bacteria;<br>• locate the human insulin gene<br>• cut/remove insulin gene (from human);<br>• cut open the plasmid;<br>• using restriction enzymes;<br>• add the human insulin gene to the plasmid;<br>• use of ligase;<br>• ref recombinant DNA;<br>• return the plasmid to a bacterial cell; | Suitably drawn **and** annotated diagrams can score full marks. | 6 |
| | | | Total 6 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| 7 (a) (i) | obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH; | | 1 |
| (ii) | sucrose; | | 1 |
| (iii) | any three from<br>• add sodium alginate with distilled water;<br>• mix (with a hand blender);<br>• until all sodium alginate has dissolved;<br>• add calcium chloride;<br>• wash beads; | | 3 |
| (b) | **B** insulin is released from the pancreas and travels to the liver<br>**A** is incorrect as glucagon does not convert glucose to glycogen<br>**C** is incorrect as glucagon is not released from the liver<br>**D** is incorrect as insulin is not released from the liver | | 1<br>] |

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 4 of 7 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| (c) (i) | • (66.65 ÷ 100) = 0.6665;<br>• (x6) = 4.0 (million); | Allow 0.67<br>Allow 3.999 | 2 |
| (ii) | • 38 49 51 62 65 78 82 91 96;<br>• 65; | | 2 |
| | | | Total 10 |

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 5 of 7 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| 8 (a) | **Advantages embryonic stem cells:**<br>• generate most cell types;<br>• can maintain and grow for long lengths of time in culture;<br>• useful for drug development studies/Parkinson's disease;<br><br>**Disadvantages of embryonic stem cells:**<br>• can cause tumours/cancer;<br>• many embryos destroyed;<br>• some people think it is unethical/killing/murder;<br>• possibility of rejection if used in transplants<br><br>**Advantages of adult stem cells:**<br>• can be reprogrammed;<br>• less likely rejected if used in transplants;<br>• taken from the patient itself;<br><br>**Disadvantages of adult stem cells:**<br>• limited ability to differentiate/develop into different types of cell;<br>• cannot be grown for long periods of time in culture;<br>• usually a very small number found in body tissues;<br>• can only grow small numbers; | Three marks from embryonic stem cell section and three marks from adult stem cell section<br><br>Allow a valid named ethic | 6 |
| | | | Total 6 |

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 6 of 12 · Question 4 part a.iii · 3 marks

2022 JUN · Paper 01 · extract 6 of 7 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| 9 (a) | • beta blockers;<br>• block/bind to receptors;<br>• which prevents the release of adrenaline/noradrenaline;<br>• heart rate decreases;<br><br>OR<br><br>• ACE inhibitors;<br>• reduce the activity of enzymes;<br>• less angiotensin;<br>• less narrowing of blood vessels; | | 4 |
| (b)(i) | any two from<br>• how much pressure blood is exerting;<br>• against artery walls;<br>• when ventricles contract/pump; | | 2 |
| (ii) | • pressure in the arteries;<br>• when the heart/ventricles at rest/relaxed/between beats; | | 2 |
| | | | Total 8 |

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 6 of 12 · Question 4 part a.iii · 3 marks

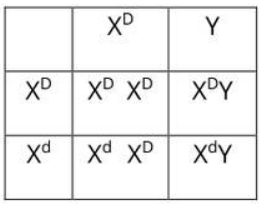
2022 JUN · Paper 01 · extract 7 of 7 · find part a.iii in this mark scheme extract

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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 7 of 12 · Question 8 part b.iii · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 8 (a) | any 4 from<br>• change in base sequence produces different mRNA;<br>• altered codon (or codons);<br>• so different amino acids coded;<br>• incorrect protein made;<br>• could produce a stop codon so no protein made/faulty protein made; | | 4 |
| (b) (i) | • faulty allele present on one chromosome only;<br>• effects overridden by a dominant allele; | | 2 |
| (ii) | • one male in generation 1 has DMD;<br>• (faulty) allele must be on X chromosome;<br>• male is XY/does not carry dominant allele on Y chromosome;<br>• carriers present; | | 3 |
| (iii) | • correct parental genotypes;<br>• correct genotypes for generation 1;<br>  | | 2 |
| | | | Total 11 |

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 7 of 12 · Question 8 part b.iii · 2 marks

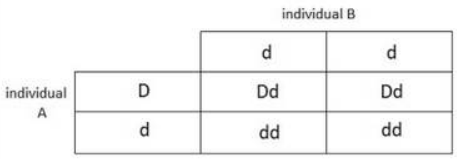
2022 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 8 of 12 · Question 7 part c.ii · 4 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 7 | (a) | Any three from:<br>• changes the DNA/base/codon sequence;<br>• changes the order of the amino acids/different amino acid coded for;<br>• changes the function/shape of the protein/different protein;<br>• affects a characteristic or function/genotype in the individual; | | 3 |
| | (b) (i) | Two from:<br>• dialysis has to be completed several times a week/transplant permanent;<br>• increased life expectancy;<br>• less restrictions on diet;<br>• better quality of life/freedom to move around/normal lifestyle; | | 2 |
| | (ii) | 100 x 3233÷4733;<br>= 68.3/68 | | 2 |
| | (iii) | Two from:<br>• people have two kidneys;<br>• two recipients from one donor;<br>• living donation/can live with just one; | | 2 |
| | (c) (i) | 0.5:1; | Accept 1:2 | 1 |
| | (ii) |  <br>• correct genotype individual A;<br>• correct genotype individual B;<br>• 50% offspring heterozygous/correct genotype and affected;<br>• 50% homozygous recessive/correct genotype and not affected; | | 4 |

Total 14 marks

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 8 of 12 · Question 7 part c.ii · 4 marks

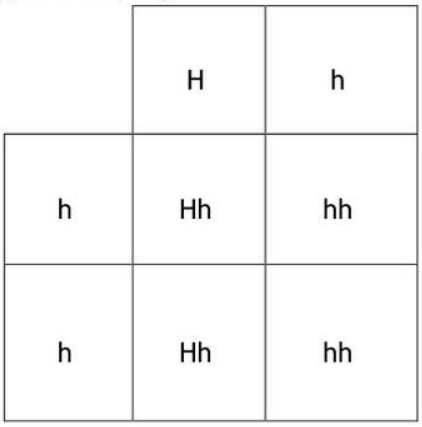
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Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 9 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

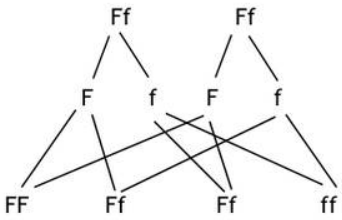
| | | | |
|---|---|---|---|
| 3 (a) | change in the order of bases in DNA; | Allow a description of a type of mutation | (1) |
| (b) | parent gametes;<br>genotypes of offspring;<br>  <br> | Allow genetic cross<br>Also allow different orientation of alleles in Punnett Square | (2) |
| (c) | **Advantages**<br>• no (costly) visits to hospital;<br>• no inconvenience of dialysis;<br>• improved quality of life/ longer life/long term solution;<br>**Disadvantages**<br>• reference to difficulty in finding a donor;<br>• rejection/reference to anti-rejection medication for life;<br>• reference to risks associated with surgery; | any one from advantages and any one from disadvantages | (2) |

**Total for Question 3 = 5 marks**

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 10 of 12 · Question 8 part a.iv · 4 marks

2024 JUN · Paper 02 · find part a.iv in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 8 | (a) (i) | **B** individual 3<br><br>The following are incorrect because:<br><br>**A** individual 1 is heterozygous<br>**C** Individual 5 is heterozygous<br>**D** individual 3 is homozygous dominant | | (1) |
| | (ii) | **D** 4<br><br>The following are incorrect because:<br><br>**A** there are more than 1 heterozygous genotypes<br>**B** there are more than 2 heterozygous genotypes<br>**C** there are more than 3 heterozygous genotypes | | (1) |
| | (iii) | • they have a dominant allele;<br>• over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ; | do not allow dominant allele overpowering the recessive allele | (2) |
| | (iv) | • heterozygous parent genotypes;<br>• gametes;<br>• genotypes of offspring;<br>• offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;<br><br>  <br> | mps 2,3 and 4 marks for correct Punnett square showing all details | (4) |
| | (b) | • reference to use of restriction enzyme<br>• (to obtain) the normal/ non faulty/ dominant allele);<br><br>• use of virus/ correctly named virus;<br>• inserts allele into person **cells**;<br>• functional/normal protein produced; | Accept liposomes<br>ignore plasmids<br><br>Penalise one for using genes instead of alleles. | (3) |

**Total for Question 8 = 11 marks**

Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

Answer 11 of 12 · Question 6 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 (a) (i) | 2500 males;<br>$\frac{2500 \times 0.8}{100}$;<br>= 20; | | ecf if use 5000 as starting figure<br>40 scores 2 marks<br>any number×0.8=1 | 3 |
| (ii) | small number/ not enough tested; | | | 1 |
| (iii) | • many more males than females with condition;<br>• show that it is sex-linked;<br>• to X chromosome;<br>• females are XX /males one X chromosome/XY<br>• females can be carriers( so fewer)/males cannot be carriers( so more);<br>• (relatively) small number (in whole population)/small chance of being colour blind;<br>• indicates that it is recessive; | | | 5 |
| (b) | parents | $X^BY \times X^BX^b$; | | 4 |
| | gametes | $X^B$ Y $X^B$ $X^b$; | | |
| | offspring genotypes | $X^B X^B$ $X^BX^b$ $X^BY$ $X^bY$; | | |
| | offspring phenotypes | normal carrier/ disease female normal female male red-green male; | If parents are incorrect ecf for mp 2,3 and 4<br><br>If X and Y are not used- max of 1 mark | |
| | | | | **Total = 13** |

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 11 of 12 · Question 6 part b · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 12 of 12 · Question 2 part b.i · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 2 | (a) | (i) | movement/swimming; | | 1 |
| | | (ii) | C;<br>A, B and D do not release energy | | 1 |
| | | (iii) | three from:<br>• produces a haploid cell / reduces the number of chromosomes by half/to 23;<br>• so at fertilisation the diploid/full number of/46 chromosomes is restored;<br>• prevents doubling of chromosomes at fertilization;<br>• introduces (genetic) variation; | | 3 |
| | (b) | (i) | sperm<br>egg<br>X / Y<br>X / XX / XY<br>X / XX / XY<br>; | One for egg<br>One for sperm<br>One for the offspring | 3 |
| | | (ii) | Two from:<br>• acrosome;<br>• contains enzymes;<br>• digests/breaks down the egg cell membrane; | | 2 |

Total 10 marks

## Type MS · Draw a genetic diagram for a monohybrid or sex-linked cross

### Answer 12 of 12 · Question 2 part b.i · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|